ELECTRONIC ANSWER BOOK

I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS, MID-TERM EXAM, AND FINAL EXAM

Instruction: Using Microsoft Word/WordPad/Notepad, type Student's name, Student ID number, type Chapter title and applicable Question Number for Assignment, Mid-Term Exam or Final Exam (as listed under Brightspace, Quizzes) before typing Solution below. One separate file for each Assignment/Mid-Term/Final Exam Question.

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Midterm – SOM – Question:

Solution:

$\alpha = 7$
S =10
M =4
a =1
$b = 0.05$
$c = -0.03$

The lateral feedback weights of the Mexican hat function are given by:

$f_{-2j} = f_{-1j} = f_{0j} = f_{1j} = f_{2j} = 0.06$ for $j = 1$ to 9
$f_{-4j} = f_{-3j} = f_{3j} = f_{4j} = -0.04$ for $j = 1$ to 9

Computation of $y_j(k)$ for $j = 1$ to 9 at any $k \geq 1$ is

$$y_j(k) = F[7 \sum_{m=-4}^{4} f_{mj} y_{j+m}(k-1)]$$

$$= F[7\{f_{-5j}y_{j-5}(k-1) + f_{-4j}y_{j-4}(k-1) + f_{-3j}y_{j-3}(k-1) + f_{-2j}y_{j-2}(k-1) + f_{-1j}y_{j-1}(k-1) + f_{0j}y_j(k-1) + f_{1j}y_{j+1}(k-1) + f_{2j}y_{j+2}(k-1) + f_{3j}y_{j+3}(k-1) + f_{4j}y_{j+4}(k-1) + f_{5j}y_{j+5}(k-1)\}]$$

For k, and $j = 1$, we have

$$\mathbf{y_1}(k) = F[7 \sum_{m=-4}^{4} f_{m,1} y_{1+m}(k-1)]$$

$$= \mathrm{F}[7 * (-0.03 * (0) - 0.03 * (0) - 0.03 * (0) + 0.05 * (0) + 0.05 * (0) + 0.05 * (0) + 0.05 * (0.1085) + 0.05 * (0.5000) - 0.03 * (0.8536) - 0.03 * (1.2800) - 0.03 * (0.8536))]$$

$$= \mathrm{F}[-0.4143]$$

$$\mathbf{y_1(k) = 0}$$

Similary solving for all

Therefore, the values of the outputs y_1 to y_{11} at iteration k are:

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
k	0	0	0.6506	1.1442	1.3596	1.1442	0.6506	0	0

And the maximum output at iteration k is $\mathbf{y_5} = 1.3596$